ANN STEFFIE MADIGO

AI Solutions Architect & AI Engineer | Enterprise AI Strategy, GenAI & Machine Learning in Banking

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Application: Head of Artificial Intelligence — Ref. HAI/IID/2026

PROFILE

AI Solutions Architect and AI Engineer in Co-operative Bank's Digital Factory and Enterprise Architecture team, where I authored the Bank's Unified AI Integration Roadmap and lead the architecture and build of its generative and agentic AI solutions. Building with technology since 2015, when I co-founded my school's computer club, and working professionally in data engineering and AI for three-plus years, with hands-on delivery across the full AI lifecycle: use-case selection, data pipelines, model development, evaluation, MLOps, security review and production hand-over. Work spans the areas this role prioritises — fraud, customer service, KYC and identity risk, and onboarding — in a CBK- and KDPA-regulated environment. I pair strategy with delivery: I write the roadmap, design the architecture, write the code, and set the governance controls the solutions run under. Completing an M.Tech in Computer Technology; AWS-certified.

LEADERSHIP & AI IMPACT — MAPPED TO THE ROLE

- **AI strategy:** Authored the Bank's Unified AI Integration Roadmap (Nov 2025), which treats AI as shared services across 9 bank products, grouped into four capability clusters (conversational AI, document intelligence, predictive analytics, personalisation), each with reusable architecture patterns and a recommended stack.
- **AI use-case delivery:** Led Rafiki, the Bank's generative-AI banking assistant (AWS Bedrock, RAG, Claude), from concept to pre-production on web and WhatsApp for the YEA/IGNITE youth segment (6M+ customers), with English/Swahili/Sheng support.
- **Fraud & identity risk:** Built ML fraud detection (K-means anomaly scoring on Kafka streams), a real-data structuring/cycling detector, a document-tamper prototype, and a face-verification ensemble for KYC, now generalised to all Kenyan ID types.
- **Responsible AI & governance:** Took Rafiki through an independent security review (21 findings) and remediation, and deferred high-risk agentic account opening to a later, controlled MVP. Identified vendor bias: a commercial KYC engine wrongly flagged 1st-generation laminated National IDs as fraud.
- **Standards & MLOps:** Set SLA-based performance-test standards (12 test categories, 111 automated cases) and database sizing approved by EA. Built CI/CD ML pipelines with experiment tracking (GitHub Actions, MLflow), and IaC for AI services (Terraform, Step Functions).
- **Value & cost:** Led the AWS Rekognition vs GBG IDscan evaluation as a lower-cost alternative for face verification; pilot reached 72.1% auto-approval at 81.2% average match confidence on records drawn from a 2.5M-record base.
- **Capability building:** Built the EA Task Tracker used by the Enterprise Architecture team (13 architects, 196+ initiatives); produce hand-over decks and design standards that let other engineers run what I build; teaching assistant for AI units at TU Kenya and at eMobilis; co-founded a school computer club teaching mobile programming.

PROFESSIONAL EXPERIENCE

Solutions Architect / AI Engineer — AI Enablement (Contract)

Aug 2025 – Present

Co-operative Bank of Kenya — Digital Factory & Enterprise Architecture, ICT & Innovation Division

Enterprise AI strategy

- Authored the Unified AI Integration Roadmap: AI as shared platform services (LLM gateway, RAG knowledge bases, document intelligence, scoring services), not one-off builds per product, covering YEA, Co-op Bank App, WhatsApp banking, CALEB, Co-op Soko, Bancassurance, Asset Finance and fraud.
- Wrote AI proposals for business owners, e.g. CALEB: OCR pre-fill, fraud/anomaly detection, predictive KYC-failure risk scoring, audit-trail intelligence and entity performance dashboards.

Rafiki — generative & agentic AI banking assistant

- Architected and built Rafiki: Next.js 15, AWS Bedrock (Claude Sonnet 4), Bedrock Knowledge Base on OpenSearch Serverless for RAG, Kafka event streaming with server-sent events, Docker, Kubernetes/ArgoCD; deployed to pre-production.
- Built a multilingual pipeline with automatic language detection and two-pass English/Swahili/Sheng translation (Bedrock ConverseStream); integrated WhatsApp Cloud API with HMAC-verified webhooks and M-Pesa STK push.
- Designed the "Rafiki AI Channel": a dedicated AI channel with its own AI Engine and WhatsApp Gateway, plus National ID + Chatbot PIN + OTP authentication for balance and account-summary use cases.
- Owned remediation of an independent security review (21 findings); scoped agentic onboarding out of the current release until controls are in place.

Identity verification & KYC (computer vision)

- Built the identity verification platform: async Python batch pipeline over ~2.5M Oracle records; serverless OCR (Textract), face comparison (Rekognition plus ArcFace/SFace ensemble), tamper detection and fraud scoring on Lambda/Step Functions (Terraform).
- Produced the vendor decision: neither GBG nor AWS alone was production-ready end to end; recommended an AWS-based verification platform. Built a passive liveness/spoof classifier, with active liveness reserved for escalations.

Jumbo Junior — birth-certificate OCR for children's accounts

- Shipped multi-rotation handling in the Bedrock OCR handler (Claude vision), now in production: when the extracted child name looks implausible and confidence is below 70%, the certificate is re-read in three orientations. Sideways captures no longer block parents, and clear photos still take a single AI call.
- Diagnosed the production timeouts this retry path caused (two sequential Bedrock calls sharing one deadline) and the image-orientation bug behind failed requests, and worked with engineering on the fix; also proposed straightening photos in the app at capture to remove the need for retries (not yet implemented).
- Fixed extraction that returned the registrar's name instead of the child's (a prompt bug), and proposed image compression and enhancement for over-size and faded captures.
- Investigated a confirmed fraud case (23+ accounts opened from one edited certificate) and prototyped tamper checks (error-level analysis, EXIF and resolution) as a proposed next control; not yet in production.

Fraud analytics

- Built a fraud-detection prototype: Django + Kafka consumers scoring transactions with K-means anomaly detection, designed to a sub-100ms latency target, with an analyst review queue.
- Now extending it to approved real transaction data to detect structuring/cycling (repeated KES 100–250 deposit–withdrawal loops on accounts opened with tampered KYC).

Solution architecture

- CALEB (staff mass account opening): led architecture: Next.js, Django, Kafka, PostgreSQL; sized production DB for 2,100 accounts/day and ~4,000 concurrent branch staff (approved by EA as-is); defined 111-case performance test suite.
- 17+ solution designs across payments (SHIF/NSSF, M-Pesa B2C Hakikisha, CRB APIs, Visa), authentication (QR passwordless), onboarding and customer experience, each with OpenAPI specs, sequence diagrams and security controls.

AWS Developer & Data Engineer

Aug 2024 – Aug 2025

A1 Strategic Business Solutions, Nairobi

- Built AWS ETL from Odoo ERP/CRM into Redshift via Glue with QuickSight dashboards, cutting monthly accounting reporting time by ~60%.
- Built an invoice-automation pipeline (Amazon Textract + Bedrock LLMs) to extract, classify and validate invoices with minimal manual work.
- Designed a Bedrock RAG conversational-AI architecture for a 7-channel banking tender (USSD, WhatsApp, web, mobile, IVR, SMS, email).

Data Engineer Intern

Jun 2023 – May 2024

ViewTech Limited / SasaPay, Nairobi

- Built mobile-money transaction data pipelines and PostgreSQL schemas for transaction processing, customer management and reporting.

AI Engineer Intern

2020 – 2022

Miradi Robotics, Nairobi

- Worked on AI and blockchain features for a home raffle management system.

MACHINE LEARNING & GENAI PROJECTS

Wellness Tourism Purchase Prediction — end-to-end MLOps | *XGBoost, scikit-learn, MLflow, GitHub Actions, Streamlit*

- CI/CD pipeline (register → prep → train) retrains on every push, tracks 144-configuration grid search in MLflow, commits the model and serves it via a live Streamlit app.
- Test set (826 customers): ROC-AUC 0.92, recall 0.79; calling the top 26% of customers reaches ~79% of buyers, at 59% conversion vs 19% baseline.
  github.com/Steffie254/wellness-tourism-purchase-prediction-mlops

NewsFindr — agentic news-briefing system | *LangGraph, LangChain, Groq LLMs, SQL agent, DuckDuckGo search*

- Agent reads a customer's interests from a SQL database, expands search queries, filters sources for credibility and writes cited summaries; redesigned the tool interface so the agent passes cache keys rather than retyping data, removing malformed tool calls; token-budget rate limiter and deterministic fallback path.

Healthcare RAG Assistant | *LangChain, Mistral-7B-Instruct (llama.cpp), Sentence Transformers, Chroma*

- Privacy-preserving, locally hosted RAG over the Merck Manual that answers clinical questions grounded in retrieved passages.
  github.com/Steffie254/healthcare-NLP-RAG-project

ReneWind — predictive maintenance deep learning | *TensorFlow/Keras, scikit-learn*

- Compared 7 neural-network architectures on 40 sensor features with class weighting and dropout; best model on the 5,000-row test set: 98.9% accuracy, 0.87 recall and 0.93 precision on the failure class.
  github.com/Steffie254/renewind-predictive-maintenance-dnn

Transaction Fraud Detection | *Django, Kafka, scikit-learn (K-means)*

- Streaming anomaly-scoring prototype with Kafka consumers and REST API; base for the real-data structuring detector above.
  github.com/AnnMadigo254/COOP-FRAUD-DETECTION

Airbnb Pricing Model | *pandas, scikit-learn, linear regression*

- Price model on log-price ($R^2 \approx 0.51$, MAPE ≈ 8.3% on test) plus EDA-driven pricing recommendations for hosts.
  github.com/Steffie254/Airbnb-ML-business-insights

MaterialsGPT Africa hackathon — 2D photocatalyst design copilot (Oct 2026, in progress) | *LLM agents, conditional VAE, graph neural network*

- Team with TU Kenya physics researchers (Track 2: design candidate materials). I lead the LLM copilot: it turns a goal such as a visible-light photocatalyst for water treatment into target properties, calls the team's CVAE generator and GNN band-gap predictor, ranks candidates and explains each pick with cited evidence; I also own the APIs and demo UI.

Nyuki — mobile text-generation app | *Django, Qwen 2.5, Flutter, Render*

- Took over a Django + Qwen 2.5 text-generation service; added a Flutter Android client and deployed to Render.

TEACHING & COMMUNITY

Teaching Assistant, AI-related units — Technical University of Kenya, 2017 – 2022
Supported lecturers and fellow students in AI-related units throughout my undergraduate studies.

Teaching Assistant — eMobilis Institute of Technology, 2017
Assisted on the Mobile Entrepreneurship Program, in which I also earned my certificate.

Co-founder, Computer Club — Kisumu Girls High School, 2015 – 2016
Co-founded the school's computer club, sponsored by eMobilis; members learned basic mobile programming in Java using IntelliJ IDEA.

EDUCATION

Master of Technology in Computer Technology (in progress) — Technical University of Kenya
Coursework: Advanced Algorithms, Research Methods, ICT Project Management, Information Security; research focus: AI-based biometric verification and KYC fraud in digital banking onboarding.

Bachelor of Technology in Computer Science — Second Class Honours (Upper Division) — TU Kenya, 2017 – 2022

CERTIFICATIONS

- **AWS Certified Data Engineer – Associate** (Aug 2025) · **AWS Certified Developer – Associate** (Dec 2024)
- Kaggle: Intro to Machine Learning, Intermediate Machine Learning, Intro to Deep Learning (2024)
- Meta Back-End Developer Professional Certificate · eMobilis Institute of Technology — Mobile Entrepreneurship Program
- In progress: AWS Solutions Architect – Associate; planned: AWS AI Practitioner → Machine Learning Specialty

TECHNICAL SKILLS

- **GenAI & agents:** AWS Bedrock (Claude), RAG, Bedrock Knowledge Bases, LangChain, LangGraph, prompt design, multilingual NLP, guardrails and evaluation
- **Machine learning:** scikit-learn, XGBoost, TensorFlow/Keras, clustering and anomaly detection, imbalanced classification, computer vision (Rekognition, ArcFace, OCR)
- **MLOps & cloud:** MLflow, GitHub Actions, Docker, Kubernetes/ArgoCD, Terraform, Lambda, Step Functions, S3, SageMaker, Glue, Redshift
- **Data & integration:** Kafka, Oracle, PostgreSQL, Redis, REST/OpenAPI, microservices, event-driven architecture, TIBCO SOA
- **Engineering:** Python (Django, Flask), TypeScript/Node (Next.js, NestJS), Vue, Java/Spring, C#
- **Governance:** CBK and Kenya Data Protection Act alignment, PCI-DSS, OAuth2, audit-trail design, human-in-the-loop decision thresholds